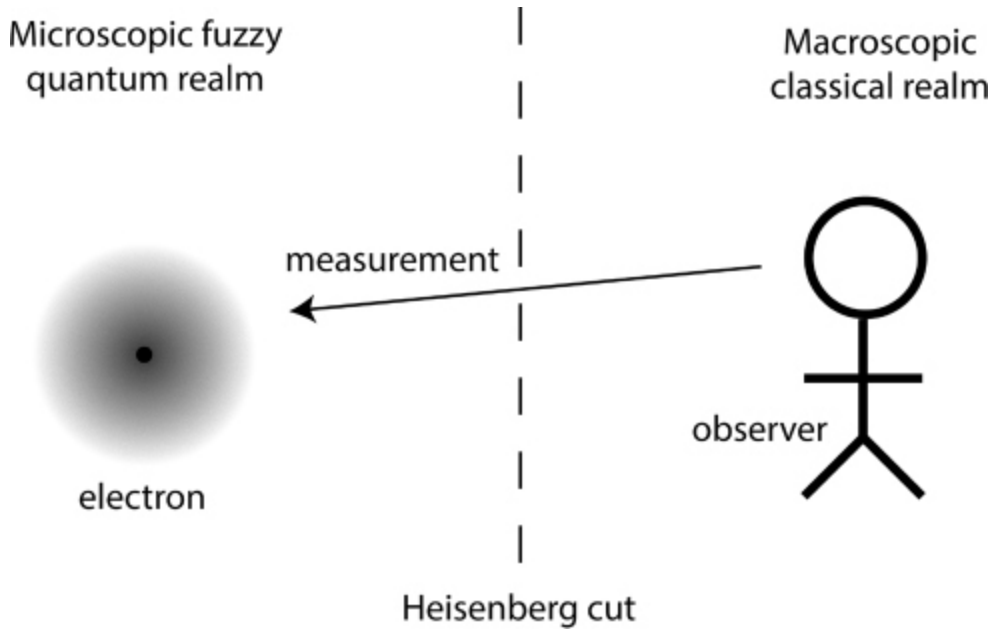


This doesn't smell right. One's first guess should be that the quantum/classical divide is a matter of our personal convenience, not a fundamental aspect of nature. If atoms obey the rules of quantum mechanics and cameras are made of atoms, presumably cameras obey the rules of quantum mechanics too. For that matter, you and I presumably obey the rules of quantum mechanics. The fact that we are big, lumbering, macroscopic objects might make classical physics a good approximation to what we are, but our first guess should be that it's really quantum from top to bottom.

If that's true, it's not just the electron that has a wave function. The camera should have a wave function of its own. So should the experimenter. Everything is quantum.

That simple shift of perspective suggests a new angle on the measurement problem. The AQM attitude is that we shouldn't treat the measurement process as anything mystical or even in need of its own set of rules; the camera and the electron simply interact with each other according to the laws of physics, just like a rock and the earth do.

A quantum state describes systems as superpositions of different measurement outcomes. The electron will, in general, start out in a superposition of various locations—all the places we could see it were we to look. The camera starts out in some wave function that might look complicated, but amounts to saying "This is a camera, and it hasn't yet looked at the electron." But then it does look at the electron, which is a physical interaction governed by the Schrödinger equation. And after that